

Table 1-1: *A Table of Our Data*

Account number	Name	Password	Balance
0	Joe	soup	100
1	Mary	nuts	3550
2	Bill	frisbee	1000
3	Sue	xyyyzz	750
4	Henry	PW	10000

The data is maintained as three global Python lists, where each list represents a column in this table. For example, as you can see from the highlighted column, all the passwords are grouped together as one list. The users' names are grouped in another list, and the balances are grouped in a third list. With this approach, to get information about one account, you need to access these lists with a common index value.

While this works, it seems extremely awkward. The data is not grouped in a logical way. For example, it doesn't seem right to keep all users' passwords together. Further, every time you add a new attribute to an account, like an address or phone number, you need to create and access another global list.

Instead, what you really want is a grouping that represents a row in the same spreadsheet, as in [Table 1-2](#).

Table 1-2: *A Table of Our Data*

Account number	Name	Password	Balance
0	Joe	soup	100
1	Mary	nuts	3550
2	Bill	frisbee	1000
3	Sue	xyyyzz	750
4	Henry	PW	10000

With this approach, each row represents the data associated with a single bank account. While this is the same data, this grouping is a much more natural way of representing an account.